

Acids, Bases and Salts

How acids and bases behave, what pH tells us, and how common salts like Plaster of Paris, baking soda and washing soda are made and used.

Introduction

Acids taste sour and change blue litmus to red. Bases taste bitter and change red litmus to blue. When an acid and a base react, they neutralise each other and form salt and water. This chapter studies these reactions as per the rationalised NCERT for 2026-27.

Common acids: HCl in stomach, acetic acid in vinegar, citric acid in lemons, oxalic acid in tomatoes. Common bases: NaOH, Ca(OH)_2 , KOH, NH_4OH .

Acids and Bases with Indicators

Indicator

An indicator is a substance that shows whether a solution is acidic or basic by a change in colour or smell.

- **Litmus:** Acid turns blue litmus red. Base turns red litmus blue. Neutral has no effect.
- **Methyl orange:** Yellow in basic, pink/red in acidic.
- **Phenolphthalein:** Pink in basic, colourless in acidic.
- **Olfactory indicators:** Onion and vanilla. Their smell changes in acid or base. Onion smell disappears in base. Vanilla smell disappears in base.

Chemical Properties

Reactions of Acids

- **With metals:** Acid + metal → salt + hydrogen. $2\text{HCl} + \text{Zn} \rightarrow \text{ZnCl}_2 + \text{H}_2$. The hydrogen burns with a pop sound.
- **With metal carbonate or metal hydrogen carbonate:** Gives salt + water + carbon dioxide. $2\text{HCl} + \text{CaCO}_3 \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$. $\text{NaHCO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$. CO_2 turns lime water milky: $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3(\text{s}) + \text{H}_2\text{O}$.
- **With bases:** Acid + base → salt + water. This is neutralisation. $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$.
- **With metal oxides:** Metal oxides are generally basic. So acid + metal oxide → salt + water. $\text{CuO} + 2\text{HCl} \rightarrow \text{CuCl}_2 + \text{H}_2\text{O}$. Black copper oxide dissolves in hydrochloric acid and forms a blue-green copper chloride solution.

Reactions of Bases

- **With metals:** Base + metal → salt + hydrogen. But only some metals do this. $2\text{NaOH} + \text{Zn} \rightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$. The product is sodium zincate.
- **With non-metal oxides:** Non-metal oxides are acidic. So base + non-metal oxide → salt + water. $2\text{NaOH} + \text{CO}_2 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$.

Strength of Acids and Bases

A strong acid or base dissociates almost fully in water. A weak acid or base dissociates only partly. HCl , H_2SO_4 , HNO_3 and NaOH , KOH are strong. Acetic acid (CH_3COOH), carbonic acid (H_2CO_3) and NH_4OH are weak.

What Acids and Bases Have in Common

All acids produce H^+ ions in water. Because H^+ cannot exist alone, it joins with water to form H_3O^+ (hydronium ion). So acids show acidic behaviour only when water is present. Dry HCl gas does not change litmus. HCl in water does.

All bases produce OH^- ions in water. Alkalis are bases that dissolve in water. All alkalis are bases, but not all bases are alkalis. Example: Cu(OH)_2 is a base but not an alkali because it does not dissolve in water.

Dilution

Mixing an acid or base with water is dilution. It lowers the concentration of H_3O^+ or OH^- per unit volume. Always add acid slowly to water with stirring. Never add water to acid, because the heat produced can cause splashing.

Worked Example 1, Why No Effect on Dry Litmus

Why does dry HCl gas not turn dry blue litmus red, but aqueous HCl does?

Dry HCl has no H^+ or H_3O^+ ions. Ions are needed to show acidic character. In water, HCl ionises: $\text{HCl} + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$. The H_3O^+ ions turn blue litmus red. So water is needed for the acid to act.

pH Scale and Its Uses

pH Scale

pH is a number that shows how acidic or basic a solution is. p in pH means potenz, which means power in German. On the pH scale from 0 to 14, pH 7 is neutral, below 7 is acidic, above 7 is basic. As pH goes down by one unit, H^+ concentration goes up by ten times.

- Acidic: 0 to 6, more H^+ ions.
- Neutral: 7, equal H^+ and OH^- .
- Basic: 8 to 14, more OH^- ions.

Universal indicator is a mix of indicators that gives different colours at different pH values and can give an approximate pH.

Importance of pH in Daily Life

- **Plants and soil:** Different plants grow best within particular soil-pH ranges. If soil is too acidic, farmers may add quicklime CaO or slaked lime $\text{Ca}(\text{OH})_2$. Organic matter can help when soil is too basic.
- **Our body:** Blood pH is about 7.4. Stomach HCl has pH around 1 to 2. During indigestion, too much acid causes pain. Antacids like milk of magnesia $\text{Mg}(\text{OH})_2$, a weak base, neutralise the excess acid.

- **Tooth decay:** After eating sugary food, bacteria produce acids. Tooth enamel begins to corrode when mouth pH falls below about 5.5. Brushing with a mildly basic toothpaste helps neutralise excess acid.
- **Bee sting and nettle sting:** Bee sting is acidic. Baking soda or mild base gives relief. Nettle sting injects methanoic acid. Rubbing with dock leaf helps.

Worked Example 2, pH Comparison

Solution A has pH 3 and solution B has pH 5. Which is more acidic?

Lower pH indicates greater acidity. Therefore, solution A is more acidic than solution B.

Salts and Chemicals from Common Salt

Salt

A salt is the product of neutralisation between an acid and a base. Family of salts means salts with the same acid or base part. Example: NaCl and Na₂SO₄ are in the sodium family. NaCl and KCl are in the chloride family.

pH of Salts

- Salt of strong acid + strong base is neutral, pH = 7. Example: NaCl.
- Salt of strong acid + weak base is acidic, pH < 7. Example: NH₄Cl.
- Salt of weak acid + strong base is basic, pH > 7. Example: Na₂CO₃, CH₃COONa.

Chemicals from Common Salt (NaCl)

Common salt is called rock salt when it comes from the ground, formed by evaporation of ancient seas. It is also made from sea water. Four useful products are made from it.

1. Sodium hydroxide (NaOH), Chlor-alkali process: When electricity is passed through brine (concentrated NaCl solution), it decomposes: $2\text{NaCl(aq)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{Cl}_2\text{(g)} + \text{H}_2\text{(g)}$. At anode: Cl₂, at cathode: H₂, in solution: NaOH. All three are useful.

2. Bleaching powder (CaOCl₂): $\text{Cl}_2 + \text{dry Ca(OH)}_2 \rightarrow \text{CaOCl}_2 + \text{H}_2\text{O}$. Preparation: $\text{Ca(OH)}_2(\text{s}) + \text{Cl}_2(\text{g}) \rightarrow \text{CaOCl}_2(\text{s}) + \text{H}_2\text{O}(\text{l})$. Uses: bleaching cotton and paper, disinfecting water, making chloroform.

3. Baking soda (NaHCO₃): Preparation: $\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 + \text{NH}_3 \rightarrow \text{NH}_4\text{Cl} + \text{NaHCO}_3$. Equation: $\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 + \text{NH}_3 \rightarrow \text{NaHCO}_3 + \text{NH}_4\text{Cl}$. Uses: antacid, baking cakes, making baking powder, fire extinguishers.

4. Washing soda (Na₂CO₃·10H₂O): From baking soda: Heating NaHCO₃ gives sodium carbonate: $2\text{NaHCO}_3 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2$. Then recrystallisation: $\text{Na}_2\text{CO}_3 + 10\text{H}_2\text{O} \rightarrow \text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$. Uses: softening hard water, washing, making glass and paper.

Baking Soda and Baking Powder

On heating, baking soda releases CO₂: $2\text{NaHCO}_3 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2$. CO₂ makes bread and cake soft and spongy. If only baking soda is used, Na₂CO₃ makes the taste bitter. So baking powder is used, which is baking soda mixed with a mild edible acid like tartaric acid. On adding water, they react and CO₂ is released without leaving a bitter residue.

Plaster of Paris (POP)

Chemical name: calcium sulphate hemihydrate. Formula: CaSO₄·1/2H₂O or 2CaSO₄·H₂O.

Preparation: Heat gypsum (CaSO₄·2H₂O) at 373 K (100 °C): $\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \rightarrow \text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O} + \frac{3}{2}\text{H}_2\text{O}$. Only half a water molecule remains per CaSO₄.

Use: When mixed with water, POP sets into hard gypsum again: $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O} + \frac{3}{2}\text{H}_2\text{O} \rightarrow \text{CaSO}_4 \cdot 2\text{H}_2\text{O}$. This is used for plastering walls, making toys and casts for broken bones, and making idols and decorative items.

Water of Crystallisation

Many salts have water molecules fixed in their crystals. This is water of crystallisation.

- Copper sulphate: blue CuSO₄·5H₂O. On heating, it loses water and becomes white anhydrous CuSO₄. On adding water again, blue colour returns. $\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \rightarrow \text{CuSO}_4 + 5\text{H}_2\text{O}$ (heat).
- Gypsum: CaSO₄·2H₂O.

- Washing soda: $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$. It has ten water molecules.

Worked Example 3, Salt Preparation

A white powder X is used by doctors on fractures. On heating, it loses water and on mixing with water it hardens. Identify X and write equations.

X is Plaster of Paris, $\text{CaSO}_4 \cdot 1/2\text{H}_2\text{O}$.

Preparation: $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (gypsum) $\rightarrow$ $\text{CaSO}_4 \cdot 1/2\text{H}_2\text{O}$ + $3/2\text{H}_2\text{O}$ at 373 K.

Setting: $\text{CaSO}_4 \cdot 1/2\text{H}_2\text{O}$ + $3/2\text{H}_2\text{O}$ $\rightarrow$ $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (hard gypsum).

Summary

- Acids give H_3O^+ in water, bases give OH^- in water.
- Acid + metal $\rightarrow$ salt + H_2 . Acid + carbonate $\rightarrow$ salt + water + CO_2 .
- pH 7 is neutral, below 7 acidic, above 7 basic. Each unit is a 10 times change in H^+ .
- pH matters for soil, digestion, tooth decay and stings.
- Salt pH depends on the strength of the acid and base that form it.
- From common salt we get NaOH, Cl_2 , H_2 , bleaching powder, baking soda and washing soda.
- Plaster of Paris is $\text{CaSO}_4 \cdot 1/2\text{H}_2\text{O}$, made from gypsum $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ at 373 K. It sets back to gypsum with water.
- Water of crystallisation is the fixed water in a salt crystal.

Practice Questions

1. What happens when you add dilute HCl to solid sodium carbonate? Give the equation and a test for the gas formed.

2. Why should you never add water to concentrated acid? What is the correct way to dilute an acid?

3. A milk of magnesia antacid has pH 10. Is it acidic or basic, and why can it help during indigestion caused by excess stomach acid?
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4. Dry HCl gas does not turn blue litmus red. Give reason.
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5. Write the preparation and two uses each of (a) bleaching powder (b) washing soda.
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6. What is water of crystallisation? Give the formulae of gypsum and Plaster of Paris and the reaction that links them.